ATOM: Authorized Tracking, Observation and Management for Intelligent Classroom Resource Management

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Abstract—**Most classrooms in educational institutions still depend on manual switching or basic sensor-based automation to manage lights, fans, air conditioners, and projectors. These setups rarely account for who is actually in the room, what kind of activity is taking place, how appliances have been behaving over time, or when a person needs to step in and take control. This paper proposes ATOM (Authorized Tracking, Observation and Management), a unified framework for intelligent classroom resource management that brings together computer vision, IoT-based control, usage-driven fault prevention, and transparent decision-making. At its core, ATOM uses a YOLOv8n-based VisionGuard module to assess classroom occupancy and identify which of the nine predefined classroom zones are in use. A Decision Engine then combines this occupancy data with predefined scene profiles and three operating modes — Automatic, Human Override, and Hybrid — to determine the most appropriate appliance configuration. FaultGuard is designed to monitor appliance behaviour through parameters such as operating duration, switching frequency, and continuous runtime, flagging unusual patterns without requiring a dedicated physical sensor on every device. ExplainAI keeps a transparent record of every automated decision, logging the timestamp, affected appliance, action taken, and the condition that triggered it. The full system is designed for implementation using an ESP32-based controller, relay modules, a USB webcam, Firebase, and a Flutter-based mobile interface. Once built, ATOM will be evaluated on how effectively it handles occupancy-aware resource control, energy management, fault prevention, and meaningful human–automation interaction.**

Index Terms—**Smart Classroom, YOLOv8n, Occupancy Detection, IoT, Energy Management, Predictive Maintenance, Explainable AI**

I. Introduction

Educational institutions represent a genuinely important area for intelligent resource management. Classroom appliances — lights, fans, air conditioners, and projectors — frequently run through the day without any real consideration of how many people are in the room or what is happening inside it. When devices operate this way, energy is consumed unnecessarily, and the problem compounds across an entire campus. IoT-based classroom systems have therefore attracted considerable research attention as a way to bring more intelligence to how these resources are managed [2], [7].

A range of approaches have been put forward for classroom automation and occupancy-aware control. IoT-based systems have been built for energy conservation, real-time monitoring, and appliance management in educational settings [1], [2], [5], [7], [8]. Camera-based methods add an important layer on top of this by revealing not just whether a room is occupied, but where people actually are. Utomo *et al.*, for instance, demonstrated a YOLOv8n-based system that adjusts classroom lighting and air conditioning in direct response to detected occupancy [10]. Camera-based occupancy positioning has also been explored more broadly for indoor energy optimisation [11]. Taken together, these studies make a compelling case for pairing richer occupancy information with automated resource control.

Occupancy-aware control has shown its value beyond classrooms as well. In HVAC systems, knowing how many people are in a space and where they are can meaningfully improve energy efficiency while keeping occupants comfortable [9]. That said, classroom automation research has largely concentrated on one function at a time — appliance control, monitoring, or energy management — and has rarely tackled equipment health monitoring or transparent explanations of why the system made a particular decision.

Predictive maintenance is another area with clear relevance here. Moat and Coleman showed that historical operational data can inform maintenance decisions even for assets without dedicated sensors [6]. Machine learning has been applied to forecast appliance failures before they occur [3], and Ferreira *et al.* explored this specifically for home appliances [4]. What these studies collectively suggest is that how an appliance is used over time carries useful diagnostic information. Integrating this kind of maintenance thinking into an active classroom resource management framework, however, remains largely unexplored.

Transparency is a dimension that tends to be underappreciated in automated systems. Explainable AI has been applied to building energy analysis to help make data-driven conclusions more interpretable and to clarify what drives consumption

[12]. Yet within classroom automation, the question of why a particular decision was made — why the lights turned off or the fan switched on — is rarely answered for the people using the space.

These observations together motivate ATOM (Authorized Tracking, Observation and Management), the unified framework proposed in this paper. ATOM integrates YOLOv8n-based occupancy detection, occupancy-aware appliance control, flexible operating modes, usage-based fault prevention, and transparent decision logging within a single coherent system. VisionGuard identifies how many occupants are present and which of the nine predefined zones they occupy. The Decision Engine uses this information alongside the current scene profile and operating mode to determine what the appliances should be doing. FaultGuard monitors usage patterns for signs of abnormal behaviour, while ExplainAI records the reasoning behind every automated action. The framework is planned for implementation using an ESP32 NodeMCU, relay modules, Firebase Realtime Database, and a USB webcam. The broader aim is to make classroom resource management smarter, more context-aware, and far easier for people to understand and trust.

II. LITERATURE SURVEY

Research on intelligent classroom management has grown steadily across several areas — occupancy detection, IoT-based automation, energy management, predictive maintenance, and explainable AI — though these areas have mostly developed in parallel rather than as parts of a unified system.

Camera-based occupancy detection has shown particular promise for adaptive classroom automation. Ani *et al.* explored an IoT-based smart classroom approach that drew on camera data to identify occupants and enable zone-level appliance control [1]. Utomo *et al.* took this further with a YOLOv8n-based system capable of adjusting classroom lighting and air conditioning based on who is detected in the room [10]. This work is especially relevant to ATOM and directly informed the design of the VisionGuard module. In a related vein, Wang *et al.* investigated camera-based indoor occupancy positioning for energy-saving applications, demonstrating that knowing where occupants are located — not just whether the room is occupied — can meaningfully support energy optimisation [11]. These studies consistently show that computer vision can offer richer occupancy information than simple presence detection, though their primary focus remains occupancy estimation and appliance or energy control.

IoT-based approaches have been equally well represented in the smart classroom literature. Challa *et al.* investigated IoT-driven energy management for classrooms [2], while Noor *et al.* presented an IoT-based energy conservation system tailored to smart classroom environments [7]. Ritheesh *et al.* focused on real-time classroom monitoring through connected devices [8], and Kumar *et al.* proposed a broader IoT-enabled smart classroom architecture aimed at improving overall classroom management [5]. These works collectively illustrate the value of connected infrastructure and remote monitoring for classroom automation. What they do not address, however, is how to combine monitoring and energy control with equipment health tracking and transparent automated decision-making.

Occupancy information has also been applied to HVAC control, where it can help strike a better balance between energy efficiency and occupant comfort. Turley *et al.* developed and evaluated an occupancy-aware HVAC control strategy for residential buildings [9]. Though not focused on classrooms specifically, this work shows clearly that adapting environmental control to actual space usage can yield meaningful improvements — a principle that directly informs the adaptive control strategy in ATOM.

Predictive maintenance offers a further important dimension for keeping classroom appliances in reliable working order. Moat and Coleman studied how historical operational data and survival analysis models can support maintenance decisions for assets that lack dedicated sensors [6]. Fernandes *et al.* applied machine learning to the problem of forecasting appliance failures [3], and Ferreira *et al.* examined predictive maintenance for home appliances with a particular focus on washing machines [4]. These studies make a persuasive case that appliance behaviour and usage history carry real diagnostic value. None of them, however, integrate this maintenance capability with occupancy-aware classroom automation.

Explainable AI is the final relevant thread in the literature. Zhang *et al.* applied interpretable models to building energy estimation, using explainable AI to surface the factors that most influence energy consumption [12]. This highlights something important: when systems make automated decisions about energy use, users benefit from being able to understand why. Yet explainability has received comparatively little attention in classroom automation research, particularly when paired with occupancy-based appliance control.

Taken as a whole, the reviewed literature confirms that computer vision can enable occupancy-aware control, IoT can support classroom monitoring and automation, predictive maintenance can improve equipment reliability, and explainable AI can make automated systems more transparent. The gap is not in any individual technology — it is in their integration. No existing study brings all of these capabilities together within a single classroom-focused framework. This is precisely the opportunity that motivates ATOM.

III. RESEARCH GAP

The literature reviewed makes clear that a considerable body of work already exists on the individual building blocks of intelligent classroom management — camera-based occupancy detection, IoT-based appliance control, occupancy-aware HVAC management, predictive maintenance, and explainable AI. What is notably absent is a framework that pulls these capabilities together into one coherent, classroom-focused system.

Camera-based approaches have demonstrated that richer occupancy information can support more adaptive energy control [10], [11], and IoT-based systems have shown how connected

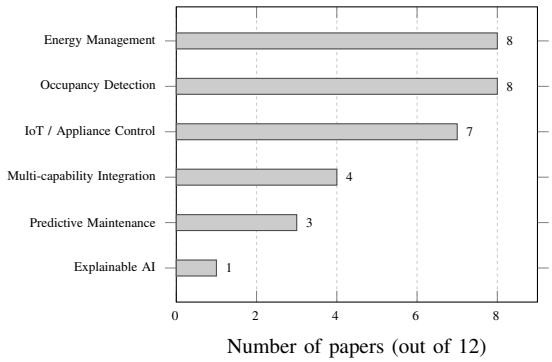


Fig. 1. Coverage of key capabilities across the 12 studies reviewed. Occupancy detection, energy management, and IoT-based control are more frequently addressed, while predictive maintenance and explainable AI receive comparatively less attention. The multi-capability category denotes studies combining multiple functions, not complete ATOM-level integration.

devices can monitor and manage classroom appliances [1], [2], [5], [7], [8]. Even so, the combination of real-time occupancy data with flexible, activity-specific control across multiple classroom resources remains largely underexplored.

Predictive maintenance has similarly been studied for facilities and appliances using historical usage data [3], [4], [6], but these efforts have not been connected to occupancy-aware classroom automation. There is therefore an open and worthwhile question: whether usage information already generated during appliance operation — things like running time and switching behaviour — can be used to flag unusual patterns within a classroom management framework, without adding a dedicated sensor to every device.

The way automated systems interact with the people using them is another gap in the literature. Existing classroom automation work largely focuses on automatic control or remote monitoring, with comparatively little thought given to designs that offer multiple operating modes and genuinely allow users to take control when needed. Similarly, while explainable AI has been explored for building energy analysis [12], transparent decision logging is not addressed alongside occupancy-based appliance control and maintenance functions in any of the reviewed studies.

Activity-specific operation is another underserved area. A regular lecture, a presentation, an examination, and an empty classroom all place meaningfully different demands on a room's resources. A system that layers real-time occupancy information over predefined activity profiles could deliver considerably more context-aware control than any single fixed automation strategy.

The core research gap, in other words, is not the absence of any particular technology. It is the lack of integration. No existing classroom system brings together occupancy detection, adaptive appliance control, usage-based fault prevention, activity-specific scene management, human override capability, and transparent decision logging in one place. This is the gap that ATOM is designed to address.

As shown in Fig. 1, the reviewed studies are heavily weighted toward occupancy detection, energy management, and IoT-based appliance control. Predictive maintenance and explainable AI appear in far fewer papers, confirming that these capabilities have received limited attention within the intelligent classroom resource management space. The pattern reinforces the case for investigating a more integrated framework — one that pairs occupancy-aware control with usage-based fault prevention and transparent decision-making.

IV. PROPOSED FRAMEWORK

Building on the gaps identified in the literature, this paper proposes ATOM (Authorized Tracking, Observation and Management), a unified framework for intelligent classroom resource management. The central idea is to bring occupancy information, classroom activity, appliance usage history, and user intervention together into a system capable of making more adaptive and context-aware resource decisions.

The architecture is organised around four main components: VisionGuard, the Decision Engine, FaultGuard, and ExplainAI. A USB webcam provides the input for occupancy detection, and a YOLOv8n-based vision module processes this feed to estimate how many occupants are present and where they are located. This information is passed to the Decision Engine, which weighs it against the current operating mode and active scene profile to produce the appropriate appliance configuration. Those control commands are then executed by an ESP32-based controller connected to the classroom appliance circuits.

Running alongside this main pipeline, appliance usage data generated during operation is fed into FaultGuard for analysis of abnormal patterns, while ExplainAI maintains a running record of every automated decision and the conditions that drove it. The full architecture is illustrated in Fig. 3.

A. *VisionGuard*

VisionGuard is the occupancy-awareness layer of ATOM. A USB webcam captures a continuous view of the classroom, and YOLOv8n processes this feed to detect and count the people present. Beyond a simple headcount, the detected positions of occupants are mapped to one of nine predefined classroom zones arranged in a 3×3 layout to determine which spatial regions are in use at any given moment. The nine-zone arrangement used for this mapping is illustrated in Fig. 2.

Crucially, this occupancy information is passed to the Decision Engine rather than being used to drive appliances directly. Keeping these steps separate means the system can factor in other considerations — the active scene profile, the current operating mode — before arriving at a control decision. An empty room, for example, might prompt unnecessary appliances to switch off, while a partially occupied classroom might allow only the resources serving the occupied zones to remain active.

B. *Decision Engine*

The Decision Engine sits at the operational centre of ATOM. It brings together occupancy data from VisionGuard,

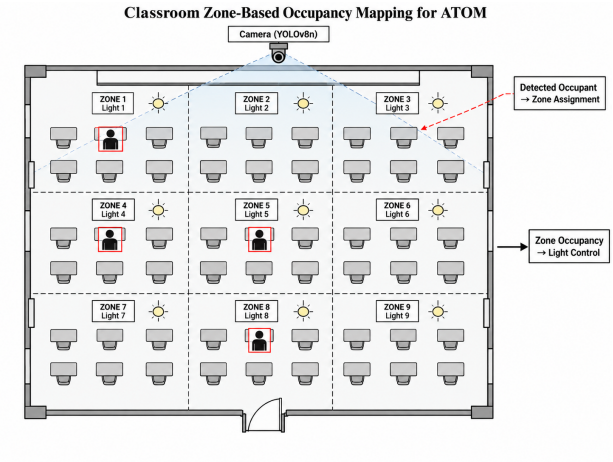


Fig. 2. Nine-zone classroom layout illustrating VisionGuard occupancy mapping for zone-aware resource control.

the selected operating mode, and the active scene profile to determine the appropriate state for each appliance.

Three operating modes are supported: Automatic, Human Override, and Hybrid. In Automatic Mode, all appliance decisions are generated from the detected classroom conditions. Human Override Mode hands direct control to an authorised user when the system's automatic judgement does not suit the situation. Hybrid Mode finds a middle ground — certain appliances can be managed manually while the rest continue operating automatically.

This structure is intentional. Classroom conditions are not always fully captured by occupancy data alone, and the system needs to accommodate situations where someone on the ground has better information than the sensors can provide. Human Override is exposed through the control interface to designated authorised users, such as the administrator or class representative, allowing direct intervention when required while preserving the automated path for normal operation.

C. Scene Profiles

To handle the variety of activities that take place in a classroom, ATOM defines four scene profiles: Normal, Presentation, Examination, and Empty. Each profile specifies a different appliance configuration baseline, applied in conjunction with real-time occupancy information.

The Normal profile covers a regular teaching session. The Presentation profile adjusts the lighting configuration when a projector is in use. The Examination profile maintains the necessary lighting and environmental conditions while keeping unnecessary equipment switched off. The Empty profile scales resource usage to a minimum when no occupants are detected.

These profiles allow the system to respond to the purpose of a session, not just the presence of people in the room.

D. FaultGuard

FaultGuard is a usage-based fault-prevention module designed to catch signs of abnormal appliance behaviour without requiring a dedicated physical sensor on every device. It works with the operational information that the control system already generates in the ordinary course of operation.

The initial design considers parameters including total operating time, switching frequency, continuous operating duration, and deviation from previously observed usage patterns. By tracking these values, the system can identify cases where an appliance is running far longer or cycling far more frequently than it typically does. When an unusual pattern is detected, the system raises a warning to prompt further inspection.

The specific prediction method, detection thresholds, and performance characteristics will be determined during the implementation and testing phase. FaultGuard is therefore presented here as a carefully reasoned proposal rather than an already-validated predictive maintenance system.

E. ExplainAI

ExplainAI is built around a straightforward but important idea: people should be able to understand why an automated system did what it did. Rather than presenting users with only the final appliance state, it maintains a structured record that captures the timestamp, the affected appliance, the action taken, and the condition responsible for that action.

A typical entry might read: "Lights switched OFF because the corresponding classroom zone is unoccupied." Equally, an action triggered by a scene profile would be linked to that profile as the recorded reason.

These decision records are intended to be stored in Firebase and accessed through the Flutter-based mobile interface. The goal is to make automated behaviour legible to the people it affects, so they can understand the system's actions and trust them — rather than simply observing the results without any context.

F. Proposed Contributions

The proposed ATOM framework focuses on bringing together the following capabilities within a single classroom resource management system:

- **Occupancy-aware resource control:** YOLOv8n-based occupancy detection provides both headcount and spatial information, enabling more precisely targeted appliance control.
- **Flexible human–automation interaction:** Automatic, Human Override, and Hybrid modes give users genuine control over when and how the system operates, without sacrificing the benefits of automation.
- **Activity-aware management:** Normal, Presentation, Examination, and Empty scene profiles allow appliance configurations to be matched to the specific activity taking place in the classroom.
- **Usage-based fault prevention:** FaultGuard analyses the operational data already produced by the system to detect abnormal behaviour, without requiring dedicated sensors on each appliance.

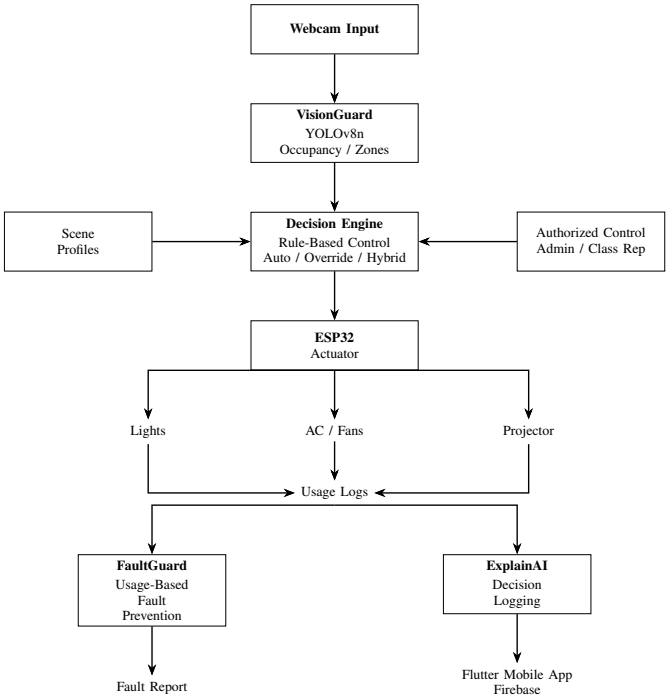


Fig. 3. Proposed architecture of the ATOM framework.

- **Transparent decision-making:** ExplainAI records automated actions alongside the conditions that triggered them, making these records accessible to users through the mobile interface.
- **Unified framework:** ATOM brings occupancy detection, appliance control, fault prevention, contextual scene operation, and decision transparency together within a single proposed classroom resource management system.

V. PROPOSED METHODOLOGY

The ATOM framework will be developed as a working prototype that integrates computer vision, embedded control, usage monitoring, and a mobile interface. A webcam will supply live classroom images to the YOLOv8n-based VisionGuard module, which identifies occupants and maps them to one of nine predefined classroom zones arranged in a 3×3 layout, as illustrated in Fig. 2. The resulting occupancy data, combined with the selected scene profile and operating mode, is processed by the Decision Engine to determine appliance states, which are then sent to the ESP32 for actuation. Appliance states and usage data are logged for FaultGuard and ExplainAI throughout operation.

A. Evaluation Metrics

The occupancy component will adopt the occupancy-detection methodology used in the YOLOv8 classroom study [10]. For non-empty test conditions, Occupancy Detection Accuracy (ODA) is computed from the actual and detected occupant counts as

$$ODA = \frac{AO - |AO - DO|}{AO} \times 100\%, \quad (1)$$

TABLE I
ATOM EVALUATION METRICS AND PLANNED MEASUREMENTS.

Component	Objective	Metrics
VisionGuard	Occupancy count	ODA, precision, recall, F1, MAE
VisionGuard	Spatial occupancy	Zone-assignment accuracy
VisionGuard	Runtime	FPS, inference latency
Decision Engine	Appliance decisions	Control accuracy, decision accuracy
Decision Engine	Responsiveness	Response latency, override success
Resource management	Utilisation	Operating time, energy (Wh), energy reduction
FaultGuard	Abnormal usage	Precision, recall, F1, false-alarm rate, lead time
ExplainAI	Transparency	Log completeness, action–reason consistency

where AO is the actual occupancy and DO is the detected occupancy. Empty-room cases will be assessed through the control confusion matrix. Precision, recall, F1-score, occupancy-count mean absolute error (MAE), zone-assignment accuracy, FPS, and inference latency will provide additional views of VisionGuard performance.

For appliance decisions, the same reference study's confusion-matrix approach [10] will be extended to lights, fans, air conditioning, and projector states under ATOM's scene profiles and operating modes. The resulting Control Accuracy is

$$Control\ Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \times 100\%. \quad (2)$$

Decision latency, override success rate, and mode-switch response time will evaluate the control interface. Resource utilisation will be measured using appliance operating time and, where reliable power measurement is available, energy consumption in Wh and percentage reduction against a conventional or fixed-schedule baseline. FaultGuard will be evaluated using precision, recall, F1-score, false-alarm rate, and detection lead time after normal and abnormal usage data are collected. ExplainAI will be assessed using decision-log completeness and action–reason consistency.

B. Experimental Protocol

The prototype will be evaluated under controlled classroom scenarios covering empty, partially occupied, and fully occupied conditions, followed by activity-specific tests for Normal, Presentation, and Examination profiles. Zone occupancy will be varied to test spatial control, while Automatic, Human Override, and Hybrid modes will be exercised separately. Additional trials with more natural occupant placement will assess robustness beyond fixed seating. For resource evaluation, ATOM will be compared with a conventional/manual or fixed-schedule baseline under common operating conditions. FaultGuard will be tested with normal usage traces and controlled abnormal patterns before predictive performance is claimed.

TABLE II
PLANNED ATOM EVALUATION SCENARIOS.

Scenario	Condition	Profile / Mode	Primary measure
Empty room	No occupants	Empty / Automatic	ODA, control accuracy
Partial occupancy	Selected zones occupied	Normal / Automatic	ODA, zone accuracy
Full occupancy	Broad zone occupancy	Normal / Automatic	ODA, latency
Presentation	Occupied + projector use	Presentation	Control accuracy
Examination	Activity-specific operation	Examination	Control accuracy
User intervention	Authorized override	Override / Hybrid	Success rate, response time
Abnormal usage	Unusual runtime/cycling	Any active profile	Fault metrics

VI. EXPECTED OUTCOMES

ATOM is expected to demonstrate that richer occupancy information can support more context-aware classroom resource control than simple presence detection. The proposed evaluation will quantify this through occupancy-count agreement, zone assignment, appliance-control accuracy, and response-time measurements rather than assumed performance claims.

The scene profiles and three operating modes are intended to provide flexibility across common classroom activities while retaining a direct intervention path for authorised users. Resource utilisation will be studied against a conventional control baseline where measurable power data are available. FaultGuard will be treated as an experimental investigation of usage-based abnormality detection, with performance determined by the quality and duration of collected usage data. ExplainAI will be assessed by whether automated actions are consistently recorded with complete and coherent reasons.

Camera visibility, lighting, occlusion, camera placement, historical usage-data availability, and reliable power measurement will define important study boundaries. Classroom video should also be handled in accordance with institutional privacy and consent requirements. The resulting measurements will determine which parts of the framework require further calibration or refinement.

VII. CONCLUSION

This paper presented ATOM (Authorized Tracking, Observation and Management), a proposed framework that combines YOLOv8n-based occupancy and zone awareness with contextual appliance control, usage-based fault prevention, authorised intervention, and transparent decision logging. Its four main components are intended to operate as a single classroom resource-management pipeline rather than isolated functions.

The paper also defines an evaluation methodology suited to the framework. Occupancy Detection Accuracy and confusion-matrix-based Control Accuracy are adapted from prior YOLOv8 classroom-control work [10], while zone accuracy, response latency, resource measurements, FaultGuard metrics, and ExplainAI log completeness extend evaluation to ATOM's additional capabilities. These metrics will support later implementation and comparison against a conventional control baseline.

The next stage is prototype implementation and structured testing. Results from those experiments will establish the practical performance, limitations, and refinement requirements of VisionGuard, the Decision Engine, FaultGuard, and ExplainAI under classroom conditions.

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